

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Scientific methodologies & sampling design - Set 1

Mark scheme

AQA spec 3.7.1; Research methods; textbook pp. 390-393

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • There is an environmental gradient (distance from the hedge / changing conditions) so samples must be taken at regular intervals along it to detect the change (1) • Random sampling might leave gaps / miss the pattern; systematic sampling avoids bias while covering the gradient (1)	2
01.2	<i>1 mark for each point.</i> • Sample at close intervals (eg every 1 m) first and see at what spacing the variations are still detected without unnecessary data; wider intervals (eg 4 m) may miss peaks and troughs (1) • Take increasing numbers of samples at one site and calculate the running mean; the number needed is where the mean stops fluctuating / stabilises (1) • Calculate standard deviation to assess scatter: more variable data need more samples (1) • Test the method (extraction technique, timing) for reliability before the main study (1)	4
01.3	<i>Up to 4 marks in total; award marks in proportion to the completeness of each point.</i> • Remove surface vegetation; pour an irritant solution (dilute mustard / detergent in water) onto the area until saturated (and around it), keep adding as it infiltrates (up to 2) • Collect worms that come to the surface for a standardised time; count / identify / weigh (up to 2) • Limitations: solution may not percolate impermeable soils or reach deep worms; worms may move sideways or downwards; some may die or not respond; slow-moving worms missed (up to 2)	4

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393; AQA spec 3.7.1.	
02.1	<i>1 mark for each point.</i> - About 8 nights (1) - The running mean fluctuates widely with few samples but stabilises at 21-22 from 8 nights onwards; extra samples do not change it significantly (1)	2
02.2	<i>1 mark for each point.</i> - Moth activity varies with temperature, wind, rain and season (adults only fly for part of the year) so catches vary independently of population size (1) - Sampling at different conditions / times would make results not comparable (1) - Timing between samples should be chosen to detect the temporal variation of interest (eg weekly through the season) (1)	3
02.3	<i>1 mark for each point.</i> - Accuracy: how close a recorded result is to the real value (1) - Precision: the smallest interval between possible recorded values (eg a ruler marked in mm is more precise than one in cm) (1)	2
02.4	<i>1 mark for each point.</i> - A null hypothesis (eg no difference / no correlation) can be tested statistically and accepted or rejected at a stated confidence level (1) - Standardised techniques ensure results collected at different times, places or by different people are comparable (1) - Other variables are controlled or monitored so their effect can be assessed; allows the study to be repeated / checked (reliability) (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393; AQA spec 3.7.1.	
03.1	<i>1 mark for each point.</i> - Trays of the same soil (texture, compaction, organic matter, depth) with different percentage vegetation cover (eg 0, 25, 50, 75, 100% grass), replicated (1) - Same slope / gradient; simulated rain from a standard height at a standard rate and drop size (eg watering can / sprinkler) for a standard time (1) - Collect the soil splashed / washed off (eg onto a sheet or into a trough at the base), dry and weigh it (1) - Repeat each treatment several times and calculate means (and standard deviation) (1) - Plot soil loss against cover; statistical test (Spearman's rank / t-test) for significance (1)	5
03.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> - Rainfall intensity / raindrop size / height (1) - Soil texture, compaction, organic matter, depth (1) - Permeability of the material beneath (1) - Gradient (1) - Root binding (1)	3

Question	Marking guidance	Marks
03.3	<i>1 mark for each point.</i> - All variables except the independent variable can be standardised in the laboratory; in the field uncontrollable variables (weather) must be measured instead (1) - Laboratory conditions may not represent natural conditions so results may not apply directly to real ecosystems (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 419-420; AQA spec 3.7.1, 3.7.2.2.	
04.1	<i>Correct answer only.</i> B (1)	1
04.2	<i>1 mark for each point.</i> - Number all possible sampling sites (eg on a grid map) (1) - Select the sites using a random number generator / table; (regular shape: grid coordinates from random numbers) (1)	2
04.3	<i>1 mark for each point.</i> - Sites may be chosen for convenience or because they appear to support (or dismiss) the hypothesis (1) - So the sample does not represent the whole area and conclusions are invalid (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 391-393; AQA spec 3.7.1.	
05.1	<i>Correct answer only.</i> The readings are precise but may not be accurate (1)	1
05.2	<i>1 mark for each point.</i> - A result that differs from other results with which it was expected to be similar (1) - Repeat the study / measurement: if it recurs it is a 'real' result caused by another variable; if not it was an error and can be excluded (1)	2
05.3	<i>1 mark for each point.</i> - To test the reliability of the method / equipment (1) - To decide the sample size, number of samples, spacing or timing needed to detect the variation without collecting unnecessary data (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-391; AQA spec 3.7.1.	
06.1	<i>1 mark for each point.</i> - About 40-50 traps (1) - Because the running mean stops changing / fluctuating with fewer than this the mean is unreliable (1)	2
06.2	<i>1 mark for each point.</i> - Widely scattered results (large standard deviation) mean a single sample may not be representative (1) - More samples are needed so that the effect of variability is averaged out and the mean is representative / significance can be assessed (1)	2

Question	Marking guidance	Marks
06.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Same type and size of pitfall trap, set for the same length of time (1) • Traps emptied at the same time of day / same weather conditions in both habitats (1) • Same number of traps and same method of positioning (random or systematic) in each habitat (1) • Other variables that could affect results (eg rainfall, temperature) monitored or controlled (1)	3
06.4	<i>1 mark for each point.</i> • Flying insect activity changes with weather over minutes to weeks (temperature, wind) and diurnally (1) • Samples on different occasions are needed to produce a mean that represents the typical value or to identify trends (1) • The interval between samples should match the rate at which the factor changes, found from a preliminary study (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 391-393; AQA spec 3.7.1.	
07.1	<i>1 mark for each point.</i> • t-test (or Mann-Whitney U if not normally distributed) (1) • Comparing two means of counted / measured data from two independent samples (1)	2
07.2	<i>1 mark for each point.</i> • The difference is statistically significant (less than 1% probability that it arose by chance) (1) • The null hypothesis (no difference between fields) is rejected; the organic field has significantly more earthworms (1)	2
07.3	<i>1 mark for each point.</i> • A significant difference does not show a causal relationship (1) • Other variables (soil type, moisture, pH, cultivation history) may differ between the two fields and cause the difference (1)	2

Question	Marking guidance	Marks
07.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Scientific methodology: identifying the topic, reviewing existing knowledge, formulating a testable hypothesis and null hypothesis, designing the study, analysing and drawing conclusions, planning further research • Sample location: random (grid coordinates / numbered sites) or systematic (transects, fixed intervals) to avoid bias; interval chosen so variation is detected without excess data • Sample timing: diurnal, seasonal, weather-related and long-term changes; sampling on different occasions; interval matched to rate of change • Sample size and number: larger and more samples for non-homogeneous variables; running mean and standard deviation show when the mean is reliable; preliminary study to decide • Standardisation and control of variables; accuracy, precision, reliability, representativeness; anomalies and repeats; calibration of equipment; appropriate techniques and risk management • Statistical analysis: choosing the test (Spearman, chi-squared, t-test, Mann-Whitney), significance and p-values, error bars, correlation vs causation • Evaluation: trade-offs between cost, time, disturbance and reliability; limitations of sub-sampling; justified conclusion	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 420-423; AQA spec 3.7.1, 3.7.2.3.	